

with DipTrace, which lets you first draw a board outline in the required shape, manually pick and place whatever parts need to be fixed and then auto-place the rest. You can then add connections and fiddle around with your design to get the right component-connected layout.

Make the most of a customisable auto-router. Before you can begin routing your design, layers, via styles and net classes need your attention. These form the foundation of your board design, and with DipTrace you can customise and group your settings. The tool gives you the freedom to decide what signals should pass through which layer, choose colours, specify hole dimensions and manage via styles and net classes.

You can then decide how the auto-router should work, thus exercising complete control over your design, effortlessly. At any stage, you may save the settings in a rules file to use in a future project.

If you are going with the auto-router for routing your wires, there are grid based and advanced shape based options. You can set it up to work the way you want and it will automatically take care of your requirements.

If you choose to do it manually, you can select from available modes and, as you route, the tool warns you of design consistency errors. Thus, you take care of the design errors as you move along. You can always edit your work and even move nets between layers.

Remaining free spaces can use the copper-pour facility for low-impedance connections and additional heat-sinking. DipTrace allows you to organise different copper pours on a single layer, as well as connecting copper polygons on different layers into a single net.

A detailed procedure is available to verify the design. This includes checking for design rules and net connectivity. With errors shown at the exact location and different types

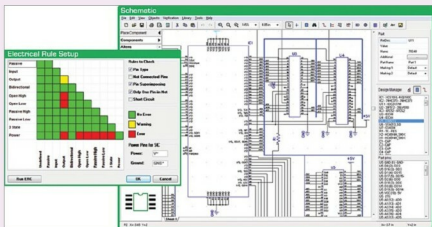


Fig. 3: Create a correct schematic with a rule check

Things you might want to know

- Offers support to Windows, Linux and MAC operating systems
- English being the primary interface language, you can change it to 21 other alternatives
- Provides free component/footprint libraries and 3D models
- If you are running it on MAC, install Xquartz beforehand
- Ace Converter and Pentalogix Viewmate are software to produce PCB using milling method and to view your Gerber, respectively
- You can purchase the proprietary version from www.diptrace.com/buy/online-store
- services@diptace.com provides PCB design, manufacturing and library creation services
- Tutorials and videos to guide you available on their website www.diptrace.com

marked in different colours, rectifying these becomes easy.

Relate the schematic and the layout. While completing your layout is important, it is as important to ensure that the schematic and the layout are in sync. There might be cases when you have to add components after the layout has been designed. With compare-to-schematic, rework-design-from-schematic and back-annotation features, this procedure is relatively simplified.

Export for manufacturing.

Gerber files for printing PCBs can be exported layer-by-layer or in batches with DipTrace. You can even specify details like what components need to be exported to the files, what accuracy is needed and preview to visualise your configuration.

Numerical control (N/C) drill file that needs to be sent in to the manufacturer can also be generated with just a few clicks. Also, with 3D-preview-&-export module, you can

see the 3D model of the board with all components installed, rotate it, change parameters to see the effect and export 3D PCB in STEP format for mechanical computer-aided design integration.

Efficient and easy to use

Having seen how you can use DipTrace to design PCBs faster and better, one might wonder how the tool manages to do what it does. Let us now take a look at three features that make the design tool work its magic.

Component editor. Manage component library, make new components and edit old ones with this editor. You can design the component, choose its properties and functioning details.

For chips, you have the choice of different kinds of packages, too. You can even group pins based on their functionality and handle these as a single unit.

Pattern editor. Every component has a particular pattern (footprint) on